

Week 2 Progress Report

Phase 2 — Iris Preprocessing and Segmentation

Objective:-

The main objective of phase 2 work is to improve the raw iris image quality so that I can use them in deep learning based iris authentication. The raw iris images are not perfect, they may contain noise, blur and reflections. So before using them in deep learning clean the images and improve the image quality.

Introduction:-

In iris recognition system image quality plays an important role in authentication. Raw iris images may contain noise, blur, reflection and extra regions that can reduce the performance.

The preprocessing can help in improving iris quality and convert the iris texture into a perfect form so that it can be easier in recognition. Some important preprocessing techniques are used in phase-2 such as Gaussian filtering, Histogram equalization, CLAHE enhancement, Segmentation and polar normalization.

Dataset Used:- CASIA-Iris-Thousand dataset was used during this phase.

Methodology:-

1) Image Resizing:-

The Raw iris image was resized to 256×256 pixels using the `cv2.resize()` function. Image resizing helps maintaining the input size fixed.

2) CLAHE Enhancement:-

CLAHE was applied to improve local contrast in the iris region. CLAHE enhancement helps make iris texture more visible without increasing image brightness.

3) Gaussian Filtering:-

Gaussian Blur filtering was applied using the `cv2.GaussianBlur()` function to reduce unwanted image noise. Noise reduction is important otherwise it may interfere with iris segmentation.

4) Histogram Equalization:-

This technique helped enhance iris texture visibility and made important iris details more visible.

5) Iris Segmentation:-

After successful circle detection, a segmentation boundary was drawn around the detected iris region. The segmentation process helped isolate the iris texture from surrounding eye regions.

6) Iris Normalization:-

Iris normalization was performed using the OpenCV `warpPolar()` function. The cropped circular iris region was transformed into a rectangular normalized representation.

7) Full Dataset Preprocessing Pipeline:-

The preprocessing pipeline was implemented on the complete data set. Preprocessing loops were created using Python to process all images in the dataset. The dataset was divided into train, validation, and test sets for future deep learning implementation. All preprocessed iris images were generated and stored in their respective folders.

Results:-

->Several Images were generated during the implementation.

- 1) Raw Iris Image:** The original iris image was successfully loaded.
- 2) Resized Iris Image:** The iris image was resized to 256×256 pixels for preprocessing
- 3) CLAHE Enhanced Image:** CLAHE enhancement improved local contrast and increased iris texture visibility.

- 4) Gaussian Blurred Image:** Gaussian filtering successfully reduced image noise.
- 5) Histogram Equalized Image:** Histogram Equalization improved overall contrast distribution and enhanced iris texture details.
- 6) Iris Segmentation Output:** The Hough circle successfully detected the iris boundary and generated segmented iris output.
- 7) Normalized Polar Iris Image:** The normalization process successfully converted the circular iris structure into a rectangular representation. Minor distortions were observed due to approximate iris boundary detection and surrounding noise regions.
- 8) Processed Dataset Generation:** Preprocessed iris images were generated and saved in train, validation, and test folders.
- 9) Dataset Split Distribution:** The dataset was successfully divided into train, validation, and test sets. A dataset distribution graph was generated.
- 10) Sample Processed Iris Outputs:** Sample processed iris images were visualized to verify preprocessing quality.

Observations:-

The preprocessing techniques improved iris image quality and texture visibility. CLAHE enhancement and histogram equalization helped improve contrast, while Gaussian filtering reduced unwanted noise.

Some normalization outputs images still contained minor distortions due to reflections, eyelids, and circle detection.

Challenges Faced:-

During this phase, problems such as incorrect image paths, OpenCV errors and wrong iris circle detection were faced. Some distortions were also observed in normalized iris outputs. However, I gradually overcame these issues by adjusting preprocessing steps, segmentation parameters.

Conclusion:-

- > I implemented preprocessing on the complete iris dataset.
- > I created train, validation, and test splits.
- > I performed image enhancement and segmentation.
- > I performed iris segmentation using Hough Circle Transform.
- > I generated normalized polar iris outputs.
- > I learned preprocessing and segmentation techniques.